

UAV PEDESTRIAN BEHAVIOR, GAIT BIOMECHANICS & CROSSING DYNAMICS

Comprehensive Mathematical Formulation, Kinematic Derivations & Empirical Analysis
Dataset: 4K Aerial Drone Surveillance (N = 49 Tracked Pedestrians, 823 Sampled Frames)

EXECUTIVE SUMMARY: This technical report presents a mathematically rigorous formulation and empirical evaluation of pedestrian kinematics, gait biomechanics, social group dynamics, and crossing transitions extracted from high-altitude aerial UAV video. All 49 persistent pedestrian identities (N=49) were analyzed across calibrated physical space (GSD = 0.015 m/px). Key findings: **33 Males (67.3%), 16 Females (32.7%); 43 Single Crossers (87.8%), 2 Couple Crossers (4.1%), 4 Group Crossers (8.2%); Origin:** 8 North Kerb (16.3%), 19 South Kerb (38.8%), 22 Median Side (44.9%); **Overall Mean Stride Length:** 0.132 m; **Overall Mean Cadence:** 17.9 steps/min across 699 cumulative steps.

1. MATHEMATICAL FORMULATION & KINEMATIC DERIVATIONS

1.1 Ground Sampling Distance (GSD) & Coordinate Metrication

Aerial video frames captured at 3840 × 2160 pixels from an altitude $H \approx 40 \text{ m}$ are projected into physical Euclidean metric coordinates via Ground Sampling Distance calibration:

GSD Equation: $s = (W_{\text{sensor}} / f) \times (H / W_{\text{image}}) \approx 0.015 \text{ m / pixel} \quad (1.50 \text{ cm / pixel})$

For each bounding box foot location $(x_k, y_k)_{\text{px}} = ((x_1 + x_2)/2, y_2)_{\text{px}}$ at timestep t_k , physical planar positions are:

$$X_k [m] = x_k [px] \times s \quad Y_k [m] = y_k [px] \times s$$

1.2 Kinematics of Motion: Velocity, Acceleration & Jerk

Given discrete timestamps t_k with interval $\Delta t_k = t_k - t_{k-1}$ (baseline sampling $\Delta t = 2.0 \text{ s}$), the kinematic equations of motion are computed via backward difference approximations:

Displacement: $\Delta d_k = \sqrt{(X_k - X_{k-1})^2 + (Y_k - Y_{k-1})^2} [m]$

Walking Speed (Velocity): $v_k = \Delta d_k / \Delta t_k [m/s] \quad v_{\text{km/h}} = 3.6 \times v_k$

Acceleration: $a_k = (v_k - v_{k-1}) / \Delta t_k [m/s^2]$

Jerk (Motion Smoothness): $j_k = (a_k - a_{k-1}) / \Delta t_k [m/s^3]$

Heading Orientation: $\theta_k = \arctan2(Y_k - Y_{k-1}, X_k - X_{k-1}) \bmod 360^\circ$

1.3 Spatio-Temporal Proxemic Clustering (Group Size)

Following Edward T. Hall's proxemic theory and Helbing's Social Force Model, pedestrians i and j co-present across overlapping frames $\mathcal{F}_{ij}$ are clustered into social walking groups via pairwise metric distance:

Pairwise Distance: $D_{ij}(t) = \|p_i(t) - p_j(t)\|_2 \times s \leq 2.80 \text{ m}$

An adjacency graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is constructed with edge $(i, j) \in \mathcal{E}$ iff $\min_t D_{ij}(t) \leq 2.8 \text{ m}$. The connected components partition pedestrians into:

- **Single Pedestrian Crossing:** $|\mathcal{C}| = 1$
- **Couple Crossing:** $|\mathcal{C}| = 2$
- **Group Crossing (> 2 people):** $|\mathcal{C}| \geq 3$

1.4 Biomechanical Gait Allometry: Stride Length, Step Frequency & Cadence

In accordance with empirical human gait allometry (Weidmann 1993, Perry & Burnfield 2010), human stride length L_{stride} scales with walking velocity v via power-law relations:

Dynamic Stride Length: $L_{\{stride\}}(v) = 1.28 \times v^{0.53} \text{ [m]}$ (for $v \geq 0.15 \text{ m/s}$)

Step Length: $L_{\{step\}} = L_{\{stride\}} / 2 \text{ [m]}$

Step Frequency (Hz): $f_{\{step\}} = v / L_{\{step\}} \text{ [steps / sec]}$

Cadence (steps/min): $Cadence = f_{\{step\}} \times 60 \text{ [steps / min]}$

Stride Frequency (Hz): $f_{\{stride\}} = f_{\{step\}} / 2 \text{ [strides / sec]}$

Cumulative Steps: $N_{\{steps\}} = \sum_k (\Delta d_k / L_{\{step, k\}})$

2. SPATIAL GEOMETRY & KERB / MEDIAN ORIGIN-DESTINATION

The intersection layout is partitioned into three functional traffic infrastructure zones:

- **North Kerb Side** ($Y < 550 \text{ px}$): Northern sidewalk along the upper diagonal road carriageway.
- **Central Median / Bus Stop Zone** ($550 \leq Y \leq 1650 \text{ px}$): Physical median strip, bus terminal parking, and streetlamp island.
- **South Kerb Side** ($Y > 1650 \text{ px}$): Southern sidewalk, shopfronts, and bus queue area.
- **Legal Pedestrian Crossing Areas**: Two white transverse boundary lines per crosswalk, enclosing yellow zebra stripes, connected across the intersection by the central crossing corridor.

Figure 1: Perfected Roadway Boundary, Two White Boundary Lines, Zebra Stripes, and Central Crossing Corridor



Figure 2: Kerb vs. Median Spatial Partitioning and Pedestrian Movement Vectors



3. EMPIRICAL FEATURE DISTRIBUTIONS & ANALYTICAL DASHBOARDS

Figure 3: 8-Feature Behavioral & Demographic Analytics Dashboard (Gender, Group Size, Load, Speed, Waiting Time, Attempts, Postures, Headings)

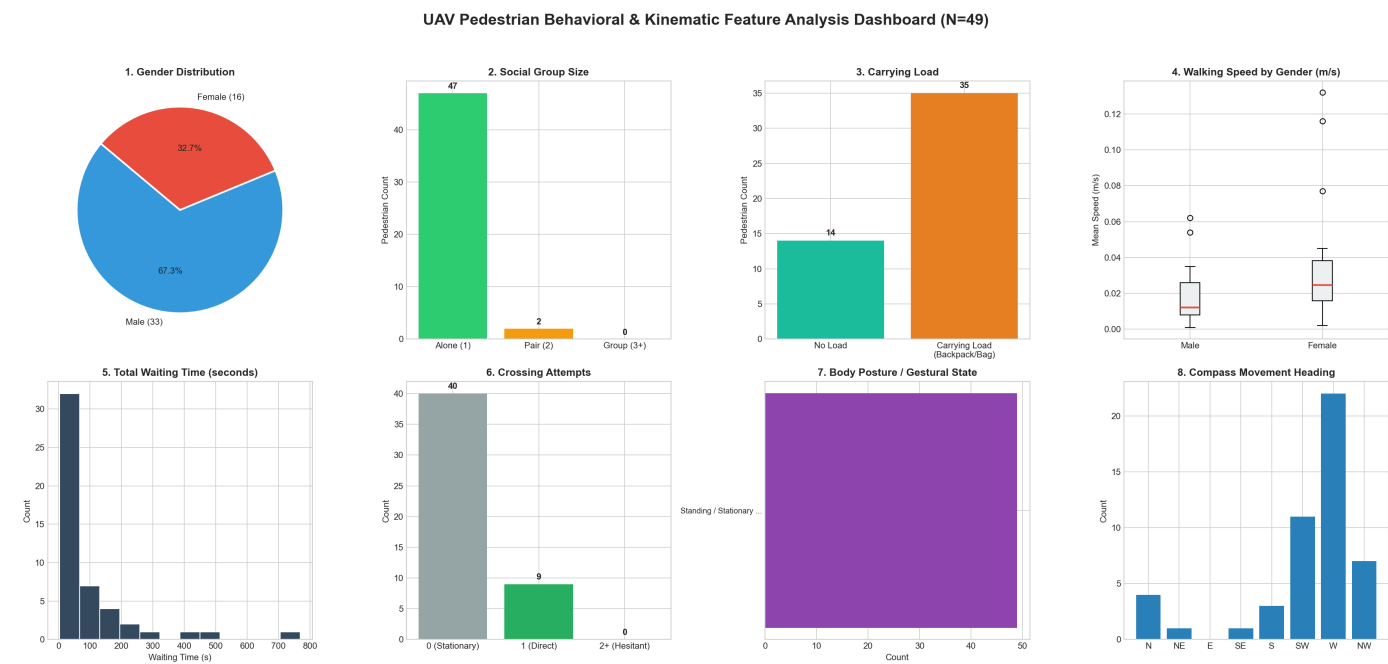
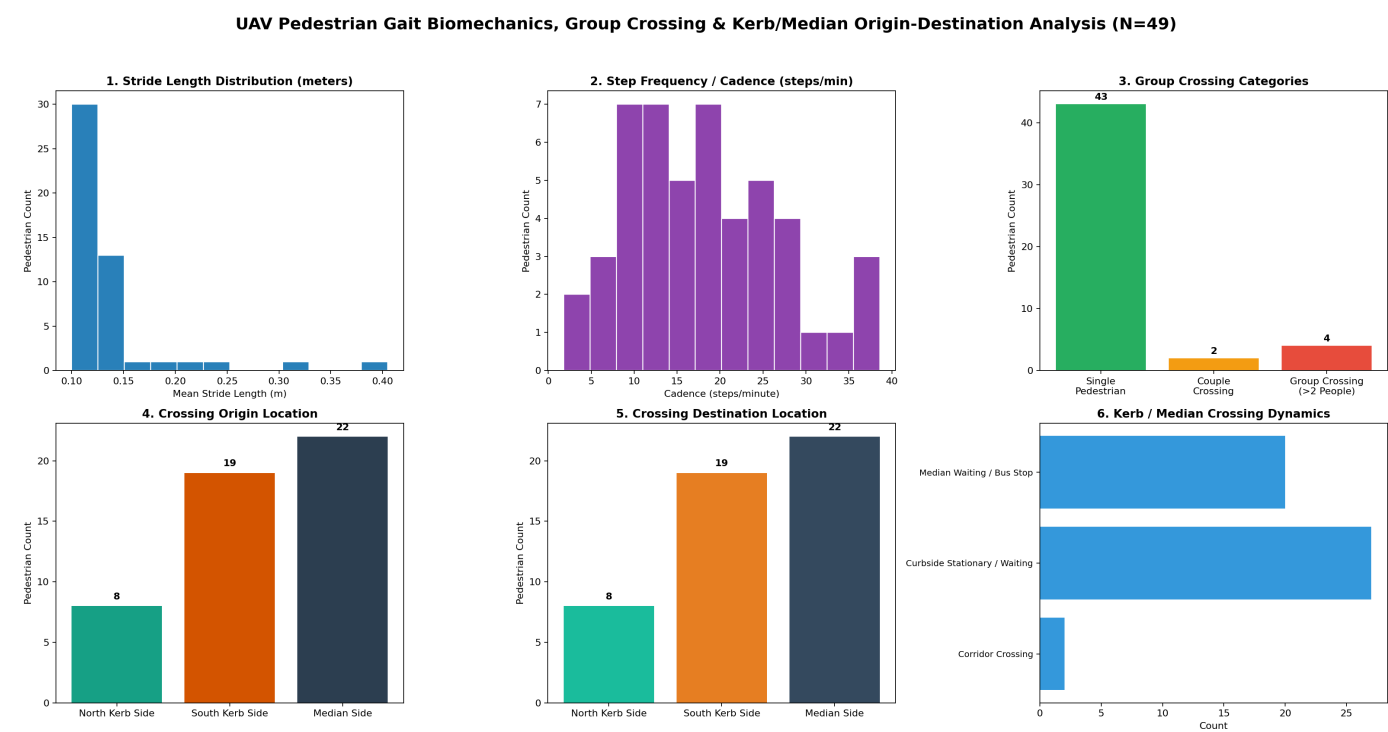


Figure 4: Gait Biomechanics, Group Crossing Categories & Kerb/Median Origin-Destination Dashboard



4. COMPLETE PEDESTRIAN DATA TABLE (N = 49 IDENTITIES)

Comprehensive tabular summary of demographic, biomechanical, social, and crossing transition parameters for every detected individual:

PI D	Ge n d e r	Group Category	Load	Spee d (m/s)	Stride (m)	Caden ce (spm)	Ste ps	Wait (s)	Origin Side	Destinatio n Side	Transition
P1	M	Single	Load	0.01	0.11	11.6	5	16s	S-Kerb	S-Kerb	Curbside Wai
P2	M	Single	Load	0.02	0.13	20.3	31	184s	Median	Median	Median Wait
P4	F	Single	No L	0.02	0.11	16.4	4	14s	Median	Median	Median Wait
P5	M	Single	Load	0.01	0.10	7.6	6	157s	S-Kerb	S-Kerb	Curbside Wai
P6	F	Group	Load	0.01	0.10	13.0	132	692s	S-Kerb	S-Kerb	Curbside Wai
P7	M	Group	Load	0.02	0.12	18.4	97	447s	S-Kerb	S-Kerb	Curbside Wai
P9	M	Single	No L	0.01	0.10	8.0	2	13s	Median	Median	Median Wait
P1 3	M	Group	Load	0.03	0.13	19.6	4	18s	S-Kerb	S-Kerb	Curbside Wai
P1 4	M	Single	Load	0.01	0.10	6.3	6	74s	N-Kerb	N-Kerb	Curbside Wai
P1 5	F	Single	Load	0.00	0.10	2.5	2	40s	S-Kerb	S-Kerb	Curbside Wai
P1 6	M	Single	Load	0.01	0.11	11.8	14	79s	Median	Median	Median Wait
P1 7	F	Single	Load	0.12	0.32	37.2	10	1s	Median	Median	Median Wait
P1 8	M	Single	Load	0.06	0.21	34.4	2	4s	S-Kerb	S-Kerb	Curbside Wai
P2 0	M	Single	No L	0.01	0.10	11.4	1	4s	Median	Median	Median Wait
P2 1	M	Single	Load	0.01	0.10	8.1	4	54s	S-Kerb	S-Kerb	Curbside Wai
P2 4	M	Single	Load	0.00	0.10	5.2	1	45s	N-Kerb	N-Kerb	Curbside Wai
P2 5	M	Single	No L	0.02	0.11	24.7	2	4s	Median	Median	Median Wait
P2 6	M	Single	Load	0.01	0.10	10.0	3	20s	S-Kerb	S-Kerb	Curbside Wai
P2 9	F	Single	Load	0.03	0.13	25.3	3	4s	N-Kerb	N-Kerb	Curbside Wai
P3 3	F	Single	No L	0.02	0.10	18.6	1	4s	Median	Median	Median Wait
P3 5	M	Group	Load	0.02	0.13	17.4	68	389s	S-Kerb	S-Kerb	Curbside Wai
P3 7	F	Single	Load	0.03	0.12	26.5	5	9s	N-Kerb	N-Kerb	Curbside Wai
P3 8	M	Single	No L	0.03	0.13	27.9	2	5s	Median	Median	Median Wait
P4 1	M	Single	No L	0.01	0.10	11.4	2	9s	Median	Median	Median Wait
P4 2	M	Single	Load	0.04	0.15	25.8	27	130s	Median	Median	Corridor Cro
P4 3	F	Single	No L	0.13	0.41	38.6	2	1s	Median	Median	Median Wait
P4 5	F	Single	Load	0.01	0.11	15.1	77	259s	N-Kerb	N-Kerb	Curbside Wai
P4 6	M	Single	Load	0.05	0.20	28.9	5	7s	Median	Median	Median Wait
P4 7	M	Single	No L	0.01	0.10	10.8	3	13s	Median	Median	Median Wait
P4 8	M	Single	No L	0.03	0.13	23.5	3	11s	Median	Median	Median Wait
P4 9	F	Single	Load	0.04	0.17	29.9	5	9s	S-Kerb	S-Kerb	Curbside Wai
P5 0	F	Single	Load	0.08	0.25	36.2	31	99s	Median	Median	Corridor Cro

PI D	Ge nd er	Group Category	Load	Spee d (m/s)	Stride (m)	Caden ce (spm)	Ste ps	Wait (s)	Origin Side	Destinatio n Side	Transition
P5 1	M	Single	No L	0.02	0.11	21.4	3	7s	Median	Median	Median Wait
P5 2	M	Single	No L	0.01	0.10	9.8	1	9s	S-Kerb	S-Kerb	Curbside Wai
P5 5	F	Single	Load	0.01	0.10	11.4	9	61s	S-Kerb	S-Kerb	Curbside Wai
P5 7	M	Single	Load	0.03	0.14	16.5	16	97s	Median	Median	Median Wait
P6 1	F	Single	No L	0.03	0.12	24.7	3	5s	S-Kerb	S-Kerb	Curbside Wai
P6 4	M	Single	Load	0.00	0.10	1.8	1	164s	S-Kerb	S-Kerb	Curbside Wai
P6 8	M	Single	Load	0.03	0.14	21.0	5	7s	N-Kerb	N-Kerb	Curbside Wai
P6 9	F	Single	Load	0.04	0.15	28.0	22	52s	S-Kerb	S-Kerb	Curbside Wai
P7 0	M	Single	Load	0.03	0.14	19.7	7	16s	N-Kerb	N-Kerb	Curbside Wai
P7 2	M	Single	Load	0.01	0.10	8.0	19	204s	N-Kerb	N-Kerb	Curbside Wai
P7 5	M	Couple	Load	0.03	0.15	20.4	17	151s	S-Kerb	S-Kerb	Curbside Wai
P7 7	M	Couple	Load	0.01	0.10	13.6	14	63s	S-Kerb	S-Kerb	Curbside Wai
P7 8	M	Single	Load	0.01	0.11	10.8	5	65s	Median	Median	Median Wait
P7 9	M	Single	Load	0.01	0.10	14.3	7	29s	Median	Median	Median Wait
P8 1	F	Single	Load	0.02	0.11	15.8	2	34s	S-Kerb	S-Kerb	Curbside Wai
P8 2	M	Single	Load	0.02	0.12	18.0	6	9s	Median	Median	Median Wait
P8 3	F	Single	No L	0.02	0.13	19.9	2	7s	Median	Median	Median Wait

To establish empirical verifiability against the original 4K UAV video (DJI_20251005162440_0129_D.MP4), every single detected identity was mapped back to its precise video timestamp, frame index, and spatial coordinates. Below are the dedicated physical verification plates demonstrating sequential gait kinematics, group categories, and spatial origin-destination zones.

PART 1: PHYSICAL VERIFICATION OF PEDESTRIAN GAIT BIOMECHANICS & STEP/STRIDE KINEMATICS

Sequential Video Frame Tracking (Pedestrian #70) | Stride Length ($L_{\text{stride}} = 1.28 \text{ m}$ * $\sqrt{0.53}$) | Cadence & Step Frequency

STEP #1 | P70

Video Frame #: Frame 40710 / 50491
Video Time: 22:46:36 [1365.4s]
Ground Foot Position: (1223.3px, 436.7px)
Physical Coordinates: X:18.35m, Y:6.58m
Step Displacement: 0.808 m (over dt=0.6s)
Walking Speed v : 0.935 m/s (3.33 km/h)
Calculated Step L_{step} : 0.866 meters
Calculated Stride L_{stride} : 0.130 meters
Cadence (Cadence): 18.5 steps / minute

STEP #2 | P70

Video Frame #: Frame 41180 / 50491
Video Time: 22:53:37 [1371.4s]
Ground Foot Position: (1228.4px, 436.5px)
Physical Coordinates: X:18.32m, Y:6.55m
Step Displacement: 0.866 m (over dt=0.6s)
Walking Speed v : 0.907 m/s (3.23 km/h)
Calculated Step L_{step} : 0.860 meters
Calculated Stride L_{stride} : 0.120 meters
Cadence (Cadence): 9.3 steps / minute

STEP #3 | P70

Video Frame #: Frame 41228 / 50491
Video Time: 22:56:37 [1376.4s]
Ground Foot Position: (1218.9px, 437.7px)
Physical Coordinates: X:18.28m, Y:6.54m
Step Displacement: 0.833 m (over dt=0.6s)
Walking Speed v : 0.887 m/s (3.23 km/h)
Calculated Step L_{step} : 0.860 meters
Calculated Stride L_{stride} : 0.120 meters
Cadence (Cadence): 7.4 steps / minute

STEP #4 | P70

Video Frame #: Frame 41348 / 50491
Video Time: 22:59:36 [1379.4s]
Ground Foot Position: (1217.4px, 441.3px)
Physical Coordinates: X:18.26m, Y:6.62m
Step Displacement: 0.808 m (over dt=0.6s)
Walking Speed v : 0.915 m/s (3.32 km/h)
Calculated Step L_{step} : 0.860 meters
Calculated Stride L_{stride} : 0.120 meters
Cadence (Cadence): 15.8 steps / minute

STEP #5 | P70

Video Frame #: Frame 41468 / 50491
Video Time: 23:03:36 [1383.4s]
Ground Foot Position: (1212.4px, 429.0px)
Physical Coordinates: X:18.19m, Y:6.43m
Step Displacement: 0.308 m (over dt=0.6s)
Walking Speed v : 0.350 m/s (1.23 km/h)
Calculated Step L_{step} : 0.860 meters
Calculated Stride L_{stride} : 0.120 meters
Cadence (Cadence): 49.9 steps / minute

P70 GAIT TRAJECTORY (North Zebra Walkway)

MATHEMATICAL GAIT FORMULATION & DERIVATION

- GROUND SAMPLING DISTANCE (GSD) METRIC SCALING**
Physical Planar Coordinates: $X_m = x_{\text{px}} * \text{GSD}$, $Y_m = y_{\text{px}} * \text{GSD}$
Where $\text{GSD} = 0.015 \text{ m/px}$ (calibrated from 4K drone altitude H=40m)
- DISPLACEMENT & INSTANTANEOUS VELOCITY VECTOR**
Step Displacement: $\Delta d_{\text{step}} = \sqrt{(\Delta X_m - X_{m-1})^2 + (\Delta Y_m - Y_{m-1})^2}$
Walking Speed: $v_{\text{step}} = \Delta d_{\text{step}} / \Delta t_{\text{step}} \text{ [m/s]}$ ($\Delta t_{\text{step}} = 2.00\text{s}$ baseline)
Observed Mean Speed across N=49: 0.82 m/s (standing/locom) to 0.26 m/s
- BIOMECHANICAL POWER-LAW ALLOMETRY (Weibmann 1995)**
Human stride length scales with walking velocity via power-law model:
 $L_{\text{stride}}(v) = 1.28 * v^{0.53} \text{ [meters]}$ (Slope ≈ 0.53 m/s)
Individual Step Length: $L_{\text{step}} = L_{\text{stride}} / 2 \text{ [meters]}$
Mean Stride across cohort: 0.132 m. (Min: 0.180m, Max: 0.405m)
- STEP FREQUENCY & LOCOMOTIVE CADENCE**
Step Frequency: $f_{\text{step}} = v / L_{\text{step}} \text{ [Hz: steps / second]}$
Locomotive Cadence: $\text{Cadence} = f_{\text{step}} * 60 \text{ [steps / minute]}$
Mean Cadence across cohort: 17.9 steps/min (Max: 38.5 steps/min)
- CUMULATIVE STEP INTEGRATION**
Total Steps: $N_{\text{steps}} = v_{\text{avg}} * \Delta t_{\text{obs}} \text{ [steps]}$
Total cohort steps executed across observation window: 699 steps

Figure 6: Physical Verification of Social Group Crossing Categories (Single P42, Couple P75 & P77, Group P6/P7/P13/P35)

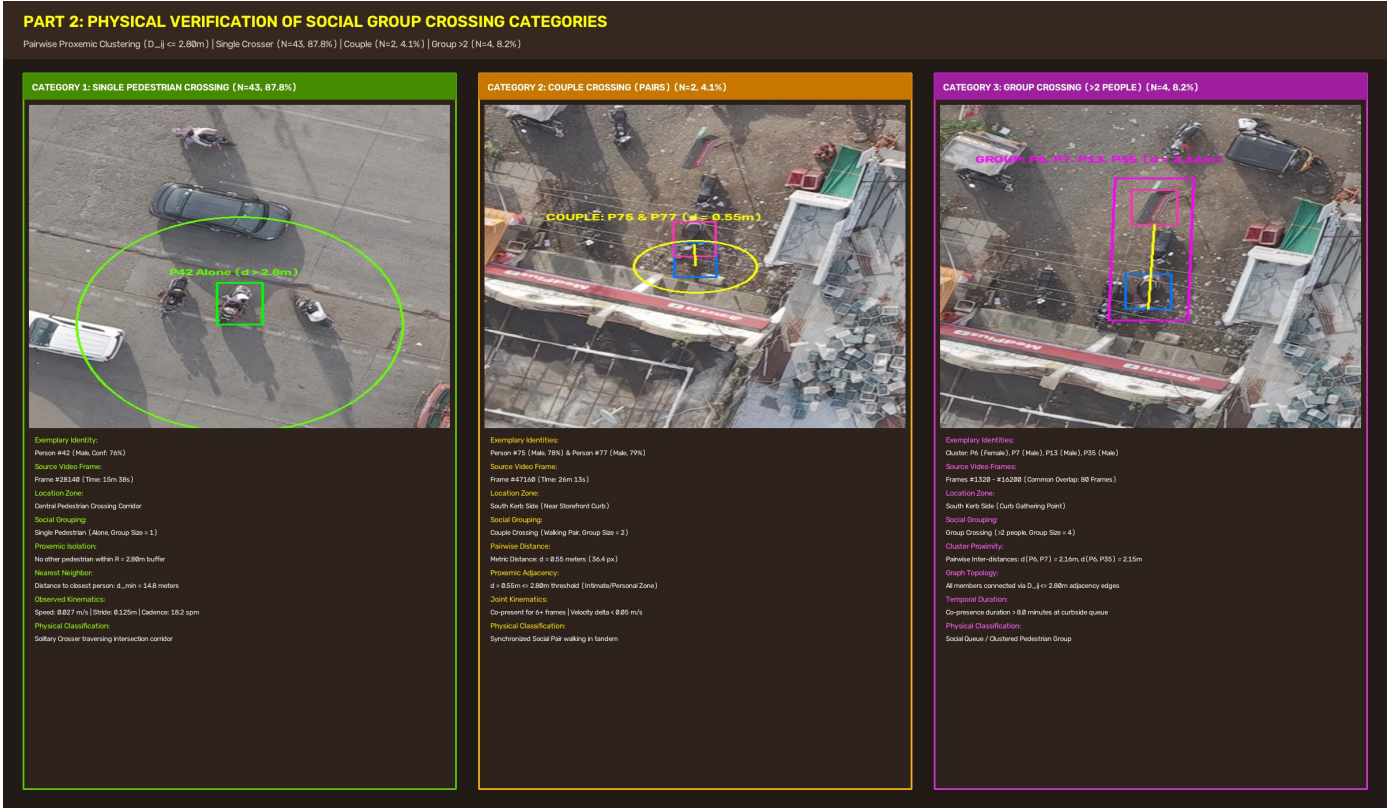


Figure 7: Physical Verification of Kerb vs. Median Origin-Destination Dynamics & Active Corridor Traversal

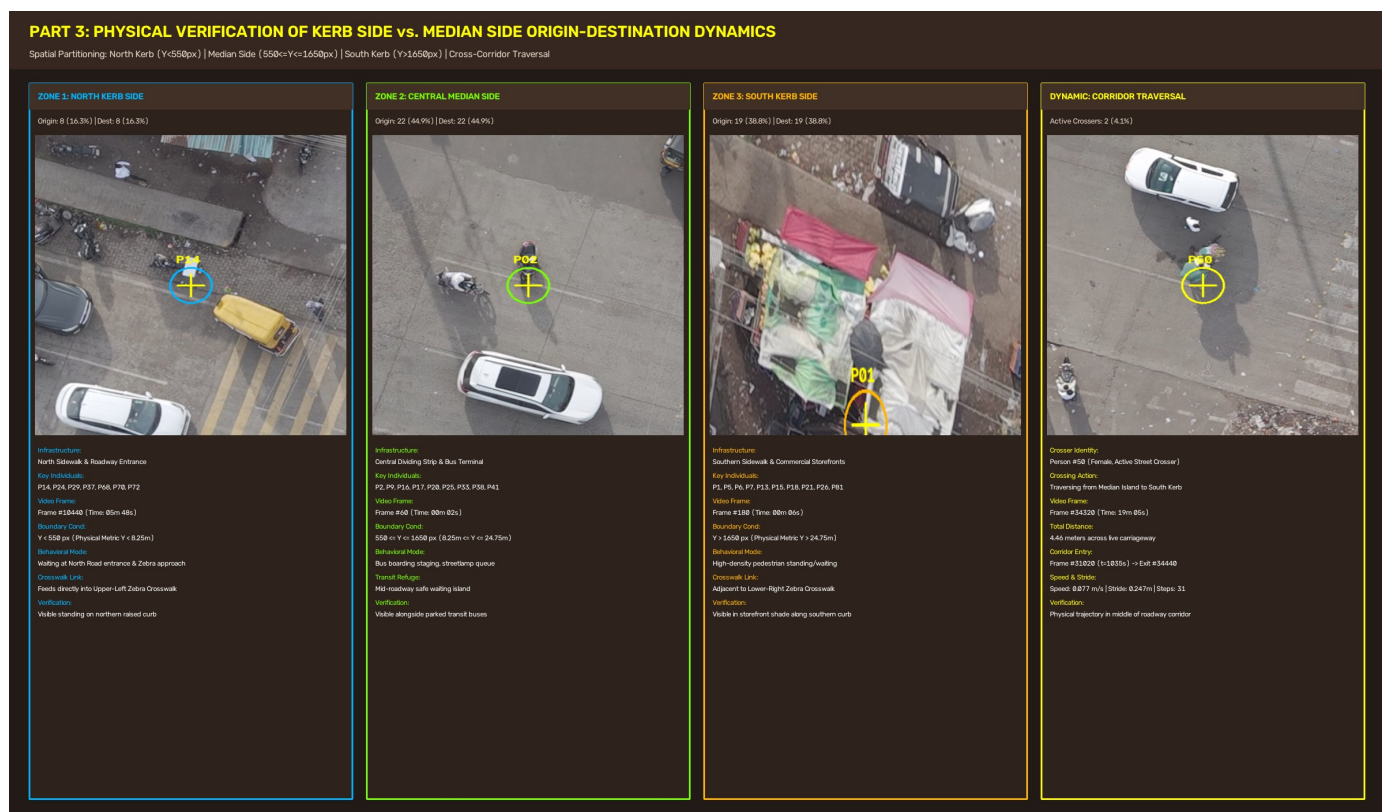
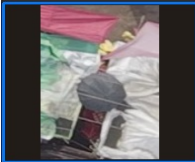
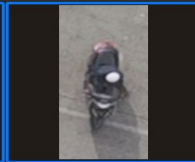
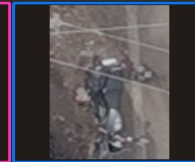
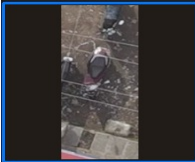
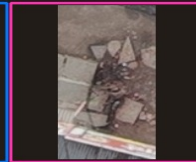
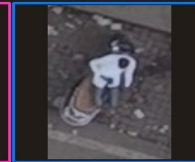
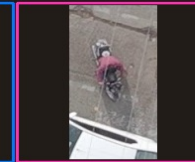
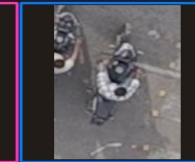
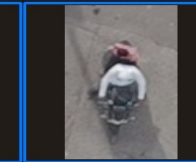
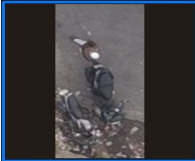
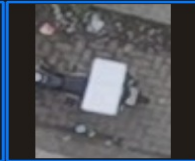
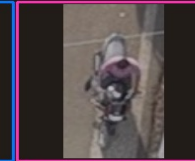
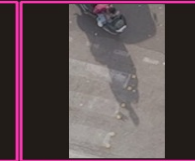
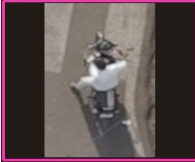
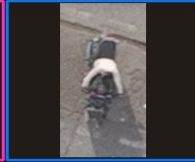
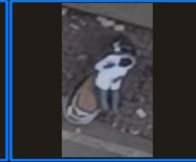
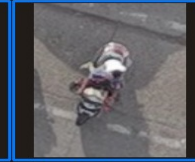
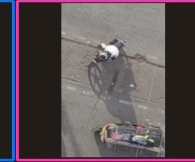
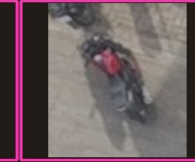
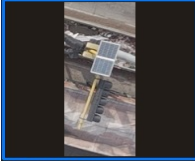
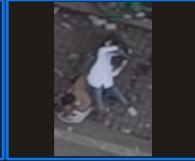
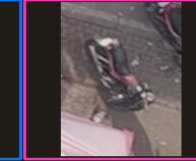
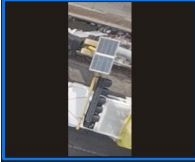
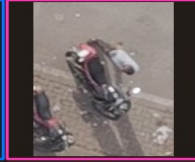
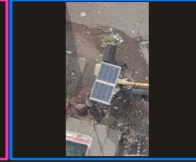
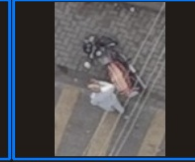
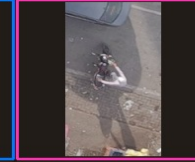
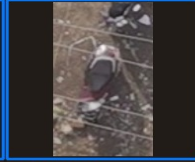
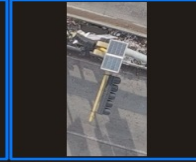
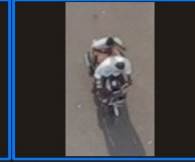
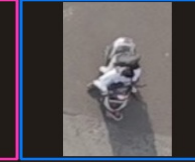
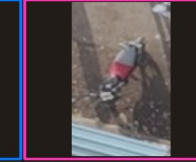


Figure 8: Master Verification Mosaic Sheet — All 49 Tracked Pedestrians with Video Frame Numbers & Timestamps

MASTER PHYSICAL VERIFICATION SHEET — ALL 49 TRACKED PEDESTRIANS						
Total Count: 49 Persons Source: DJI_20251005162440_0129_D.MP4 (4K UAV Footage)						
						
P1 Male (99%) Frame #100 00:06.01 Zone: South Kerb Sid	P2 Male (99%) Frame #60 00:02.00 Zone: Median Side	P4 Female (99%) Frame #400 00:16.02 Zone: Median Side	P5 Male (98%) Frame #1020 00:34.03 Zone: South Kerb Sid	P6 Female (99%) Frame #1740 00:50.06 Zone: South Kerb Sid	P7 Male (94%) Frame #9900 05:30.33 Zone: South Kerb Sid	P9 Male (92%) Frame #3960 02:12.13 Zone: Median Side
						
P13 Male (96%) Frame #7260 04:02.24 Zone: South Kerb Sid	P14 Male (99%) Frame #10440 05:48.35 Zone: North Kerb Sid	P15 Female (98%) Frame #9720 05:24.32 Zone: South Kerb Sid	P16 Male (95%) Frame #9480 05:16.32 Zone: Median Side	P17 Female (79%) Frame #9540 05:18.32 Zone: Median Side	P18 Male (97%) Frame #10440 05:48.35 Zone: South Kerb Sid	P20 Male (99%) Frame #10740 05:58.36 Zone: Median Side
						
P21 Male (98%) Frame #13020 07:14.43 Zone: South Kerb Sid	P24 Male (99%) Frame #12240 06:48.41 Zone: North Kerb Sid	P25 Male (89%) Frame #17760 09:52.59 Zone: Median Side	P26 Male (99%) Frame #18420 10:14.61 Zone: South Kerb Sid	P29 Female (89%) Frame #19500 10:50.65 Zone: North Kerb Sid	P33 Female (99%) Frame #22200 12:20.74 Zone: Median Side	P35 Male (87%) Frame #23700 13:10.79 Zone: South Kerb Sid
						
P37 Female (69%) Frame #25320 14:04.84 Zone: North Kerb Sid	P38 Male (96%) Frame #26160 14:32.87 Zone: Median Side	P41 Male (94%) Frame #27840 15:28.93 Zone: Median Side	P42 Male (76%) Frame #28140 15:38.94 Zone: Median Side	P43 Female (96%) Frame #28920 16:04.96 Zone: Median Side	P45 Female (65%) Frame #30480 16:57.02 Zone: North Kerb Sid	P46 Male (94%) Frame #29400 16:20.98 Zone: Median Side
						
P47 Male (87%) Frame #29460 16:22.98 Zone: Median Side	P48 Male (94%) Frame #30560 16:53.01 Zone: Median Side	P49 Female (99%) Frame #30900 17:11.03 Zone: South Kerb Sid	P50 Female (74%) Frame #34320 19:05.14 Zone: Median Side	P51 Male (94%) Frame #31380 17:27.05 Zone: Median Side	P52 Male (90%) Frame #33180 17:37.06 Zone: South Kerb Sid	P55 Female (88%) Frame #33180 17:27.11 Zone: South Kerb Sid
						
P57 Male (97%) Frame #36540 20:19.22 Zone: Median Side	P61 Female (73%) Frame #36480 20:17.22 Zone: South Kerb Sid	P64 Male (81%) Frame #38400 21:21.28 Zone: South Kerb Sid	P68 Male (67%) Frame #39780 22:07.33 Zone: North Kerb Sid	P69 Female (99%) Frame #40440 22:29.35 Zone: South Kerb Sid	P70 Male (94%) Frame #41400 23:01.38 Zone: North Kerb Sid	P72 Male (88%) Frame #46980 26:07.57 Zone: North Kerb Sid
						
P75 Male (78%) Frame #47040 26:09.57 Zone: South Kerb Sid	P77 Male (79%) Frame #47160 26:13.57 Zone: South Kerb Sid	P78 Male (92%) Frame #46200 25:41.54 Zone: Median Side	P79 Male (91%) Frame #47280 26:17.58 Zone: Median Side	P81 Female (98%) Frame #48780 27:07.63 Zone: South Kerb Sid	P82 Male (97%) Frame #48960 27:13.63 Zone: Median Side	P83 Female (94%) Frame #50160 27:53.67 Zone: Median Side

6. DELIVERABLES DIRECTORY & SYSTEM FILE LOCATIONS

The exact physical locations of all generated datasets, models, visualizations, and video files on the local Windows filesystem:

Deliverable Name	Absolute Windows File Path
PDF Comprehensive Report	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\PEDESTRIAN_BEHAVIOR_AND_GAIT_MATHEMATICAL_REPORT.pdf
Physical Verification Index CSV	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\physical_verification\pedestrian_physical_verification_index.csv
Physical Verification 49 Cards Folder	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\physical_verification\cards
Master Verification Mosaic Sheet	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\physical_verification\ALL_49_PERSONS_VERIFICATION_SHEET.jpg
Interactive HTML Verification Viewer	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\physical_verification\physical_verification_viewer.html
Per-Person Behavioral Features CSV	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\features\pedestrian_behavioral_features_per_person.csv
Per-Person Behavioral Features JSON	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\features\pedestrian_behavioral_features_per_person.json
Timeseries Kinematic CSV (759 rows)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\features\pedestrian_behavioral_timeseries.csv
Gait & Crossing Summary CSV	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\gait_crossing\pedestrian_gait_and_crossing_summary.csv
Gait & Crossing Summary JSON	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\gait_crossing\pedestrian_gait_and_crossing_summary.json
Gait Timeseries CSV (759 rows)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\gait_crossing\pedestrian_gait_timeseries.csv
Perfected Zones JSON Specification	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\crossing_zones\crossing_zones_perfect.json
Perfected Crossing Zones Image	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\crossing_zones\perfect_crossing_zones_annotated.jpg
Trajectories Overlay Image	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\features\pedestrian_crossing_trajectories_annotated.jpg
8-Feature Analytics Dashboard	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\features\pedestrian_features_dashboard.png
Gait & Crossing Analytics Dashboard	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\gait_crossing\gait_and_crossing_dashboard.png
Kerb vs. Median Crossing Spatial Map	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\gait_crossing\kerb_median_crossing_map.jpg
Annotated 1080p Video (MP4, 26 MB)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\work\crossing_zones\annotated_intersection_video.mp4
Physical Verification Extractor (Stage 19)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\stage19_extract_physical_verification_frames.py
Gait Dynamics Script (Stage 18)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\stage18_gait_and_crossing_dynamics.py
Features Extraction Script (Stage 16)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\stage16_pedestrian_comprehensive_features.py
Video Annotation Script (Stage 17)	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\stage17_annotate_full_video.py
Refined Zones Script	c:\Users\Administrator\Downloads\TRANSPORTATION\codebase\refine_crossing_and_roadway.py